RobotReviewer can be supportive in Risk of Bias Assessment in Nursing RCTs

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Background and objective

The online tool RobotReviewer enables an automated risk of bias assessment using machine learning (Marshall et al., 2016). An evaluation study including 1180 RCTs from different health topics yielded a Cohen's kappa agreement between .10 and .48 comparing RobotReviewer and human assessment, a sensitivity between .28 and .76, and a specificity between .72 and .90 for detecting a low risk in selection, performance, and detection bias (Gates et al., 2018). The performance of the RobotReviewer in nursing-related RCTs is unclear. Therefore, the objective of this study was to evaluate the reliability of RobotReviewer's risk of bias assessment in nursing-related RCTs.

Method

Research design: Evaluation study

- Index test: RoB via RobotReviewer
- Reference test: RoB reported in Cochrane reviews

Literature search process

Cochrane reviews with nurs* in title were identified in MEDLINE via PubMed on August 30th, 2018

Inclusion criteria

Electronical English language full text RCTs

Data extraction and assessment

Two independent research teams (JH/JM & PS/GM)

Results

The selection process yielded 190 RCTs published between 1959 and 2016 in 23 Cochrane reviews published between 2000 and 2018. Missing assessments of risk of bias domains in Cochrane reviews or RobotReviewer yielded varying sample sizes per risk of bias domain.

Sensitivity ranged from .44 to .88 and specificity from .48 to .95. Positive predictive value was highest for allocation concealment (.79) and lowest for blinding assessors (.25). Cohen's Kappa was moderate for randomization (.52), allocation concealment (.60), and for blinding of personal/patients (.43). Blinding of outcome assessors had only slight agreement (.04) (Table 1).

Table 1: Robot Reviewer's performance

Domain (n)	Sn (95%-CI)	Sp (95%-CI)	PPV (95%-CI)	NPV (95%-CI)	Kappa (95%-CI)	Percentage observed agreement
Randomization (129)	.88 (.81; .95)	.62 (.48; .75)	.77 (.69; .86)	.78 (.69; .87)	.52 (.36; .68)	.78
Allocation concealment (180)	.77 (.68; .86)	.82 (.75; .90)	.79 (.7; .88)	.81 (.72; .89)	.60 (.48; .72)	.80
Blinding of personal/patients (109)	.44 (.19; .68)	.95 (.90; .99)	.58 (.3; .86)	.91 (.74; 1.07)	.43 (.14; .72)	.87
Blinding of outcome assessors (115)	.58 (.39; .77)	.48 (.38; .59)	0.25 (1.4; .35)	.80 (.7; .9)	.04 (-1.14; .22)	.50

Abbreviations: NPV=Negative predictive value; PPV=Positive predictive value; Sn=Sensitivity; Sp=Specificity

Conclusions

This first evaluation of the RobotReviewer's performance in nursing-related RCTs yielded moderate agreement with the human assessment of Cochrane reviews for randomization and allocation concealment as well as an adequate sensitivity for detecting low risk of selection bias. RobotReviewer's performance might be better with good reported RCTs and the use of consistent wording. Based on these results, the use of RobotReviewer for the risk of bias assessment in nursing-related RCTs can be supportive but should be supervised by human assessment.

Download, correspondence, and declarations

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All authors declare that they have no conflict of interest.